# NETCAD YAZILIM A.Ş.

CASE STUDY REPORT



HACETTEPE UNIVERSITY
GEOMATICS ENGINEERING



# MERHABA!

Gizem CAN

# SUNUM İÇERİĞİ

1.API'den gelen verileri PostgreSQL veritabanına aktarma

2. ShapeFile dosyasını PostgreSQL veritabanına aktarma

3. DXF dosyasını PostgreSQL veritabanına aktarma

4. Excel dosyasını PostgreSQL veritabanına aktarma

5. Tüm tablolara bir kolon açılarak bu kolona ilgili İlçe bilgilerini aktarma

6. Nöbetçi Eczaneler verisinin İlçelerin merkez noktalarına olan uzaklarını hesapma

7. Başvuran Kişi verilerinin Nöbetçi Eczanelere olan uzaklıklarını hesaplama

Bonus



# API’den Gelen Verileri PostgreSQL Veritabanına Aktarma

Q1-1.py × bonus.py

Q1-1.py > ...

```python
import http.client
import psycopg2
import json

# PostgreSQL veritabanı bağlantısı
conn_postgres = psycopg2.connect(
    database="eczane",
    user="postgres",
    password="123456",
    host="localhost",
    port= 5432
)
cur_postgres = conn_postgres.cursor()

# API bağlantısı
conn_api = http.client.HTTPSConnection("api.collectapi.com")

headers = {
    'content-type': "application/json",
    'authorization': "apikey 4qq5VVSdgJppXDX3ik5ftZ:1BGnPkR5jho5bK3CKplnCS"
}

conn_api.request("GET", "/health/dutyPharmacy?ilce=%C3%87ankaya&il=Ankara", headers=headers)

res = conn_api.getresponse()

data = res.read()
```

```
pip install jsonlib
pip install psycopg2
pip install httpclient
```

Veritabanı üzerinde sorguları çalıştırmak ve sonuçları almak için kullanılan bir araçtır.

API'ya GET isteği gönderilir ve API'den gelen yanıt (response) okunur.

```python
# API'den gelen veriyi JSON olarak çözümle
api_data = json.loads(data.decode("utf-8"))
# Verileri PostgreSQL veritabanına eklemek
for pharmacy in api_data["result"]:
    name = pharmacy.get("name")
    dist = pharmacy.get("dist")
    address = pharmacy.get("address")
    phone = pharmacy.get("phone")
    loc = pharmacy.get("loc")

    # Verinin benzersiz bir özelliği ile kontrol yap
    cur_postgres.execute(
        "SELECT * FROM pharmaciess WHERE name = %s AND address = %s",
        (name, address)
    )
    existing_pharmacy = cur_postgres.fetchone()

    # Veri yoksa eklemeyi yap
    if not existing_pharmacy:
        cur_postgres.execute(
            "INSERT INTO pharmaciess (name, dist, address, phone, loc) VALUES (%s, %s, %s, %s, %s)",
            (name, dist, address, phone, loc)
        )
        conn_postgres.commit()
        print(f"{name} eklendi.")
    else:
        print(f"{name} zaten var, eklenmedi.")

print("Veri aktarım işlemi tamamlandı.")
```

CollectAPI

Shell | Go | Node | Javascript | Java | Python | Ruby | Csharp | Swift | Ocaml | Php

```
curl --request GET \
  --url 'https://api.collectapi.com/health/dutyPharmacy?ilce=%C3%87ankaya&il=Ankara' \
  --header 'authorization: apikey 4VaSd6lMNcPPAqbcoyDWwM:6x5PJwSgjU27cj8MbCVzpJ' \
  --header 'content-type: application/json'
```

Response


```
{
  "success": true,
  "result": [
    {
      "name": "Sevinç Eczanesi",
      "dist": "Çankaya",
      "address": "Maltepe Mah. Güzaltan Sk. No:1/C Maltepe",
      "phone": "0312 231 71 75",
      "loc": "39.92887565500589,32.84444332122803"
    },
    {
      "name": "Alper Yüce Eczanesi",
      "dist": "Çankaya",
      "address": "Angora Cad. No:154/Beysukent",
      "phone": "0312 236 36 96",
      "loc": "39.88566860058295,32.71559000015259"
    },
    "..."
  ]
}
```


| API Name | Package | Date | Status |
|---|---|---|---|
| Nöbetçi Eczane API | Free | 2023-08-20 | Active |

CollectAPI

Query | Query History

```
1  SELECT * FROM public.pharmaciess
2
```

Data Output | Messages | Notifications

| | name text | dist text | address text | phone text | loc text |
|---|---|---|---|---|---|
| 1 | YEŞİLBAĞ | ÇANKAYA | Mutlukent Mah. 2074. Sok. No:11 | 3122354884 | 39.899314,32.707654 |
| 2 | YAŞAMKENT MİNE | ÇANKAYA | ALACAATLI MAH. MEHMET RİFAT BÖREKÇİ CAD. NO:35/B-E | 3125024050 | 39.853974,32.647384 |
| 3 | ROYAL | ÇANKAYA | Bahçelievler Mah.Kazakistan Cad. No:5/D ÇANKAYA/ANKA... | 3122154770 | 39.931859,32.823205 |
| 4 | YENİ BAYRAMOĞLU | ÇANKAYA | MARMARA SOK. NO:20/A SIHHIYE | 3124356969 | 39.926599,32.85885 |
| 5 | MİRAY | ÇANKAYA | ÖVEÇLER MAH. AHMET HASIM CAD. NO:80/A | 3124836321 | 39.88692198,32.83328016 |
| 6 | YENİ ÇUKURAMBAR | ÇANKAYA | KIZILIRMAK MAH. 1425. CAD. NO:33/A ÇANKAYA | 3122844688 | 39.901246,32.812006 |
| 7 | DÖRT MEVSİM | ÇANKAYA | BİRLİK MAH. 401.CADDE NO:14/B | 3124959305 | 39.883546,32.875025 |



# İlçelere ait ShapeFile Dosyasını PostgreSQL Veritabanına Aktarma

test2.py > ...

```python
from sqlalchemy import create_engine, text
from geoalchemy2 import Geometry
import geopandas as gpd

user = "postgres"
password = "123456"
host = "localhost"
port = 5432
database = "eczane"

conn = f"postgresql://{user}:{password}@{host}:{port}/{database}"
engine = create_engine(conn)

# Read shapefile using GeoPandas
shapefile_path = 'C:/Users/user/Desktop/CaseStudy/CaseStudy/İlçeler/ilceler_ankara_3857.shp'
gdf = gpd.read_file(shapefile_path)

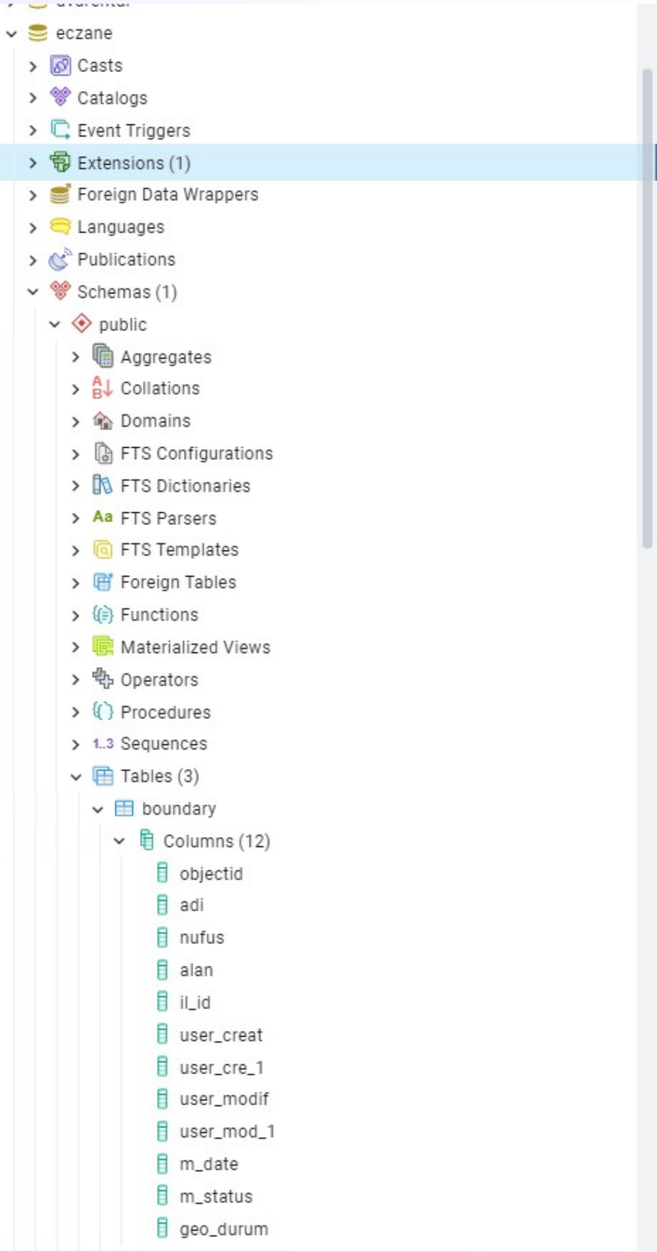
# Define the geometry column type using Geometry from geoalchemy2
geometry_type = Geometry(geometry_type='GEOMETRY')

# Drop the geometry column temporarily
gdf_no_geom = gdf.drop(columns=['geometry'])

# Import the DataFrame without geometry to the database
gdf_no_geom.to_sql(name="boundary", con=engine, schema="public", if_exists="replace", index=False)

# Add the geometry column with the correct type using GeoAlchemy2
# Add the geometry column with the correct type using GeoAlchemy2
with engine.connect() as connection:
    connection.execute(text(f"ALTER TABLE public.boundary ADD COLUMN geometry {geometry_type};"))
    connection.execute(text(f"UPDATE geometry_columns SET srid=3857 WHERE f_table_name='boundary' AND f_geometry_column='geometry';"))
    for index, row in gdf.iterrows():
        connection.execute(text(f"UPDATE public.boundary SET geometry = ST_GeomFromText(:geom, 3857) WHERE index = :index;"), geom=row['geometry'].w

print("success")
```




public.boundary/eczane/postgres@PostgreSQL 15

No limit

Query | Query History

```
1  SELECT * FROM public.boundary
2
```

Data Output | Messages | Notifications

| | objectid bigint | adi text | nufus bigint | alan double precision | il_id bigint | user_creat double precision | user_cre_1 double precision | user_modif double precision | user_mod_1 double precision | m_date double precision | m_status double precision | geo_durum bigint |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 20 | NALLIHAN | 29209 | 2064562771.97561 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 2 | 21 | POLATLI | 121858 | 3417652064.52202 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 3 | 2 | ALTINDAĞ | 363687 | 158087685.60773 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 4 | 4 | BALA | 21618 | 1838805217.77147 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 5 | 7 | ÇANKAYA | 922536 | 405053298.575768 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 6 | 8 | ÇUBUK | 86055 | 1153958210.78625 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 7 | 18 | KIZILCAHAMAM | 25179 | 1628363973.99857 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 8 | 11 | EVREN | 2847 | 218142462.355786 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 9 | 12 | GÖLBAŞI | 122288 | 1500824326.10754 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 10 | 10 | ETİMESGUT | 527959 | 149998118.843283 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 11 | 19 | MAMAK | 607878 | 342333031.698939 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 12 | 6 | ÇAMLIDERE | 6479 | 753671444.29128 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 13 | 17 | KEÇİÖREN | 889876 | 154241081.372811 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 14 | 25 | YENİMAHALLE | 632286 | 420410475.517265 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 15 | 23 | SİNCAN | 506950 | 877338619.330992 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 16 | 1 | AKYURT | 30245 | 372164960.912904 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 17 | 3 | AYAŞ | 12678 | 1148105710.1146 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 18 | 5 | BEYPAZARI | 47582 | 1642328417.80295 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 19 | 9 | ELMADAĞ | 43776 | 632285857.778195 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 20 | 13 | GÜDÜL | 8392 | 515471095.945902 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 21 | 14 | HAYMANA | 28355 | 2200469299.95409 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 22 | 15 | KAZAN | 51764 | 590328711.044076 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 23 | 16 | KALECİK | 13388 | 1115667656.39241 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 24 | 22 | PURSAKLAR | 133961 | 151221564.220721 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |
| 25 | 24 | ŞEREFLİKOÇHİSAR | 33729 | 2045185937.56817 | 1 | [null] | [null] | [null] | [null] | [null] | [null] | 1 |



# Maden Sınırlarına Ait  DXF Dosyasını PostgreSQL Veritabanına Aktarma

```python
import ezdxf
import psycopg2
```

pip install ezdxf pip install psycopg2

```python
# PostgreSQL bağlantısı
conn = psycopg2.connect(
    dbname="eczane",
    user="postgres",
    password="123456",
    host="localhost",
    port= 5432
)
cursor = conn.cursor()

# DXF dosyasını açma
dxf_file = 'C:/Users/user/Desktop/CaseStudy/CaseStudy/Maden Sınırları/maden_sinirlari.dxf'
doc = ezdxf.readfile(dxf_file)

# DXF data insertion into PostgreSQL
for entity in doc.modelspace().query('*'):
    layer = entity.dxf.layer
    paperspace = entity.dxf.paperspace
    linetype = entity.dxf.linetype
    entityhandle = entity.dxf.handle

    if entity.dxftype() == 'TEXT':
        text = entity.dxf.text
    elif entity.dxftype() == 'MTEXT':
        text = entity.plain_text()
    elif entity.dxftype() == 'ATTRIB':
        text = entity.dxf.text
    else:
        text = None  # Default to None if entity type doesn't have text

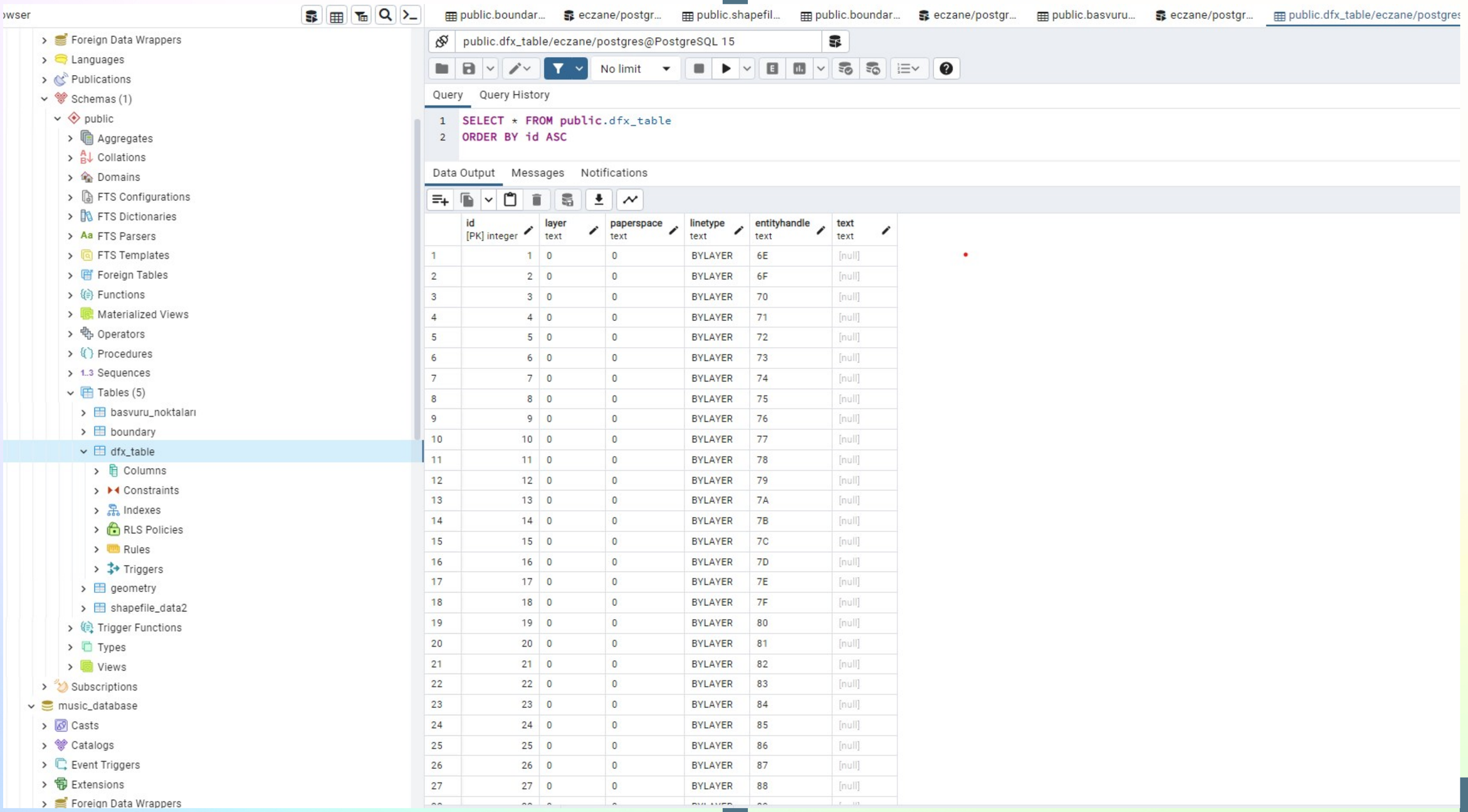
    # PostgreSQL insertion query
    insert_query = """
    INSERT INTO dfx_table (layer, paperspace, linetype, entityhandle, text)
    VALUES (%s, %s, %s, %s, %s)
    """
    values = (layer, paperspace, linetype, entityhandle, text)

    cursor.execute(insert_query, values)

# Finalize the process
conn.commit()
cursor.close()
conn.close()
```

DXF Verilerinin Döngü İçinde Okunması

Varlık Türüne Göre Metin Verisinin Çıkarılması


public.boundar...
eczane/postgr...
public.shapefil...
public.boundar...
eczane/postgr...
public.basvuru...
eczane/postgr...
public.dfx_table/eczane/postgres
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (5)
basvuru_noktaları
boundary
dfx_table
Columns
Constraints
Indexes
RLS Policies
Rules
Triggers
geometry
shapefile_data2
Trigger Functions
Types
Views
Subscriptions
music_database
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
public.dfx_table/eczane/postgres@PostgreSQL 15
No limit
Query
Query History
1 SELECT * FROM public.dfx_table
2 ORDER BY id ASC
Data Output
Messages
Notifications
id [PK] integer
layer text
paperspace text
linetype text
entityhandle text
text text
1 1 0 0 BYLAYER 6E [null]
2 2 0 0 BYLAYER 6F [null]
3 3 0 0 BYLAYER 70 [null]
4 4 0 0 BYLAYER 71 [null]
5 5 0 0 BYLAYER 72 [null]
6 6 0 0 BYLAYER 73 [null]
7 7 0 0 BYLAYER 74 [null]
8 8 0 0 BYLAYER 75 [null]
9 9 0 0 BYLAYER 76 [null]
10 10 0 0 BYLAYER 77 [null]
11 11 0 0 BYLAYER 78 [null]
12 12 0 0 BYLAYER 79 [null]
13 13 0 0 BYLAYER 7A [null]
14 14 0 0 BYLAYER 7B [null]
15 15 0 0 BYLAYER 7C [null]
16 16 0 0 BYLAYER 7D [null]
17 17 0 0 BYLAYER 7E [null]
18 18 0 0 BYLAYER 7F [null]
19 19 0 0 BYLAYER 80 [null]
20 20 0 0 BYLAYER 81 [null]
21 21 0 0 BYLAYER 82 [null]
22 22 0 0 BYLAYER 83 [null]
23 23 0 0 BYLAYER 84 [null]
24 24 0 0 BYLAYER 85 [null]
25 25 0 0 BYLAYER 86 [null]
26 26 0 0 BYLAYER 87 [null]
27 27 0 0 BYLAYER 88 [null]



# Başvuru Noktaları Excel dosyasını PostgreSQL Veritabanına Aktarma

ROJECT
db.py
Q3.py
Q4.py
test2.py
test4.py

Q4.py > ...

```python
import openpyxl
import psycopg2

# Excel dosyasını oku
wb = openpyxl.load_workbook('C:/Users/user/Desktop/CaseStudy/CaseStudy/basvuru_noktaları.xlsx')
sheet = wb.active

data = []
for index, row in enumerate(sheet.iter_rows(values_only=True)):
    if index == 0:  # Skip header row
        continue
    data.append(row)

# PostgreSQL veritabanına bağlan
conn = psycopg2.connect(
    dbname="eczane",
    user="postgres",
    password="123456",
    host="localhost",
    port=5432
)
cursor = conn.cursor()

for row in data:
    sql = "INSERT INTO basvuru_noktaları (basvuru_numarasi, kisi_adi, adresi, sehri) VALUES (%s, %s, %s, %s)"
    cursor.execute(sql, row)

# Değişiklikleri kaydet ve bağlantıyı kapat
conn.commit()
cursor.close()
conn.close()

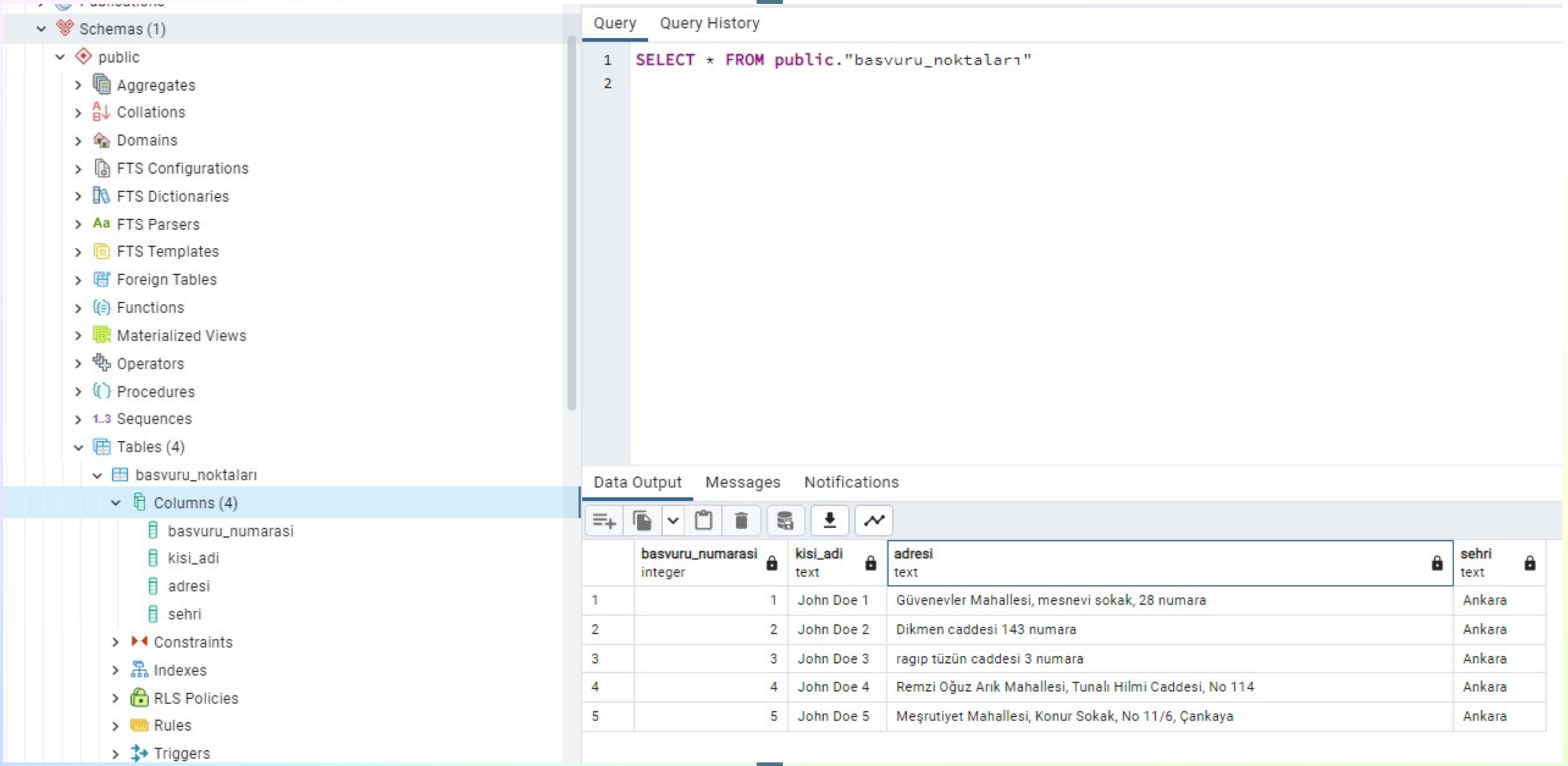
print("Veriler PostgreSQL'e aktarıldı.")
```

```
pip install openpyxl pip install psycopg2
```

Verilerin Okunması ve Saklanması

Verilerin data Listesine Eklenmesi

Önceki adımlarda okunan ve data listesine saklanan Excel verilerini, bir PostgreSQL veritabanına eklemek


Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (4)
basvuru_noktaları
Columns (4)
basvuru_numarasi
kisi_adi
adresi
sehri
Constraints
Indexes
RLS Policies
Rules
Triggers
Query
Query History
1 SELECT * FROM public."basvuru_noktaları"
2
Data Output
Messages
Notifications
basvuru_numarasi integer
kisi_adi text
adresi text
sehri text
1 1 John Doe 1 Güvenevler Mahallesi, mesnevi sokak, 28 numara Ankara
2 2 John Doe 2 Dikmen caddesi 143 numara Ankara
3 3 John Doe 3 ragıp tüzün caddesi 3 numara Ankara
4 4 John Doe 4 Remzi Oğuz Arık Mahallesi, Tunalı Hilmi Caddesi, No 114 Ankara
5 5 John Doe 5 Meşrutiyet Mahallesi, Konur Sokak, No 11/6, Çankaya Ankara



# Tüm tablolara Bir Kolon Açılarak Bu Kolona İlgili İlçe'nin Bilgilerini Aktarma

Query  Query History

```sql
-- dfx_table tablosuna ilce_id sütunu ekleme
ALTER TABLE dfx_table
ADD COLUMN ilce_id integer;

-- basvuru_noktaları tablosuna ilce_id sütunu ekleme
ALTER TABLE basvuru_noktaları
ADD COLUMN ilce_id integer;

-- pharmaciess tablosuna ilce_id sütunu ekleme
ALTER TABLE pharmaciess
ADD COLUMN ilce_id integer;

```

```sql
UPDATE basvuru_noktaları
SET ilce_adı = CASE
    WHEN basvuru_numarasi = 1 THEN 'MAMAK'
    WHEN basvuru_numarasi = 2 THEN 'ÇANKAYA'
    WHEN basvuru_numarasi = 3 THEN 'YENİMAHALLE'
    WHEN basvuru_numarasi = 4 THEN 'ÇANKAYA'
    WHEN basvuru_numarasi = 5 THEN 'ÇANKAYA'
    ELSE ilce_adı
END;
```

ery Query History

```
SELECT * FROM public."basvuru_noktaları"
```

ta Output Messages Notifications

| basvuru_numarasi integer | kisi_adi text | adresi text | sehri text | ilce_id integer | ilce_adı text | location text |
|---|---|---|---|---|---|---|
| 1 | John Doe 1 | Güvenevler Mahallesi, mesnevi sokak, 28 numara | Ankara | 19 | MAMAK | 39.895529890677054, 32.8540612362797 |
| 2 | John Doe 2 | Dikmen caddesi 143 numara | Ankara | 7 | ÇANKAYA | 39.891974665659134, 32.8417267261978 |
| 3 | John Doe 3 | ragıp tüzün caddesi 3 numara | Ankara | 25 | YENİMAHALLE | 39.96048921855266, 32.81581416852985 |
| 4 | John Doe 4 | Remzi Oğuz Arık Mahallesi, Tunalı Hilmi Caddesi, No 1… | Ankara | 7 | ÇANKAYA | 39.902884397819015, 32.8602389685280 |

Query Query History

```
1  SELECT * FROM public.pharmaciess
2
```

Data Output Messages Notifications

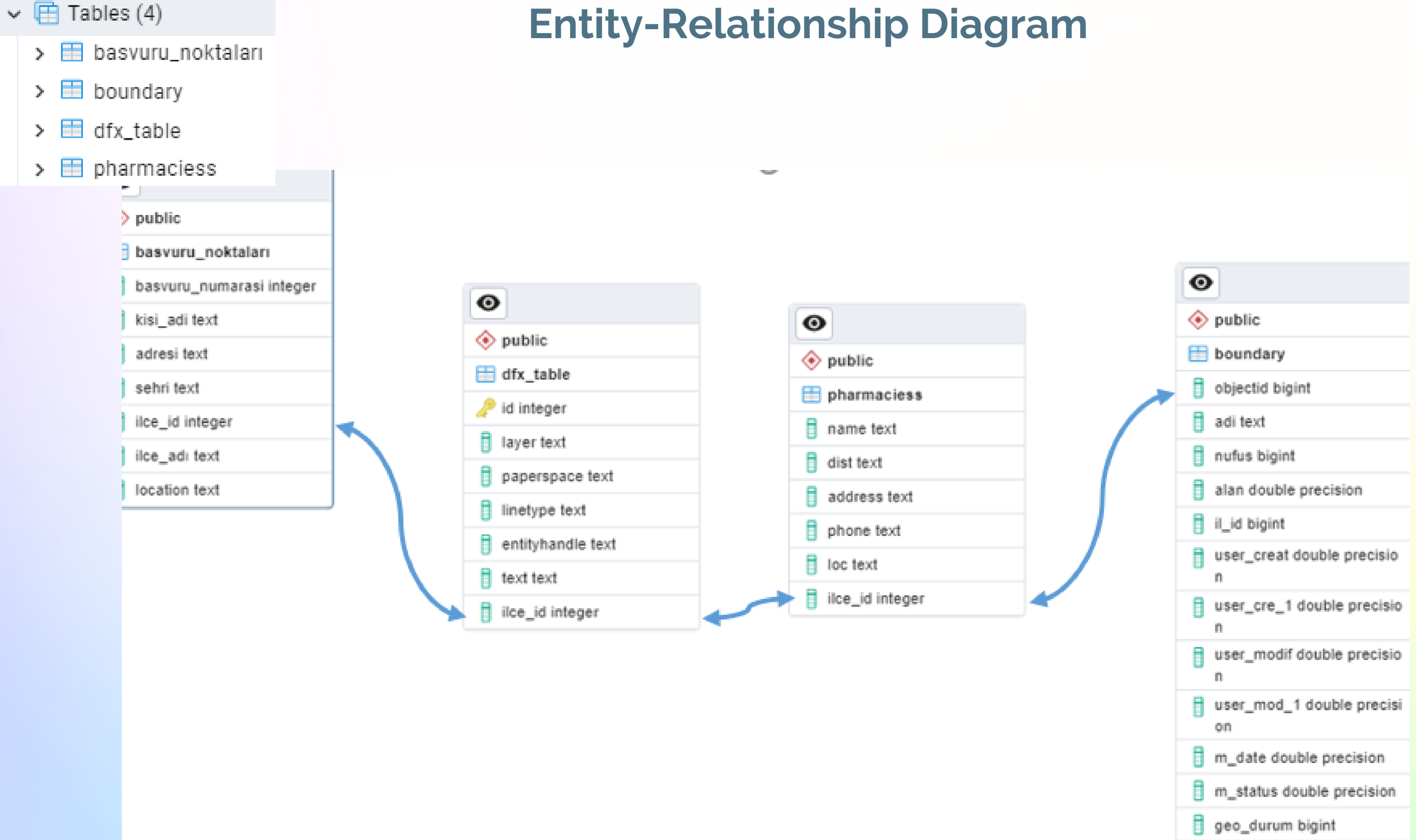
| | name text | dist text | address text | phone text | loc text | ilce_id integer |
|---|---|---|---|---|---|---|
| 1 | YEŞİLBAĞ | ÇANKAYA | Mutlukent Mah. 2074. Sok. No:11 | 3122354884 | 39.899314,32.707654 | 7 |
| 2 | YAŞAMKENT MİNE | ÇANKAYA | ALACAATLI MAH. MEHMET RİFAT BÖREKÇİ CAD. NO:35/B-E | 3125024050 | 39.853974,32.647384 | 7 |
| 3 | ROYAL | ÇANKAYA | Bahçelievler Mah.Kazakistan Cad. No:5/D ÇANKAYA/ANKA… | 3122154770 | 39.931859,32.823205 | 7 |
| 4 | YENİ BAYRAMOĞLU | ÇANKAYA | MARMARA SOK. NO:20/A SIHHIYE | 3124356969 | 39.926599,32.85885 | 7 |
| 5 | MİRAY | ÇANKAYA | ÖVEÇLER MAH. AHMET HASIM CAD. NO:80/A | 3124836321 | 39.88692198,32.83328016 | 7 |
| 6 | YENİ ÇUKURAMBAR | ÇANKAYA | KIZILIRMAK MAH. 1425. CAD. NO:33/A ÇANKAYA | 3122844688 | 39.901246,32.812006 | 7 |
| 7 | DÖRT MEVSİM | ÇANKAYA | BİRLİK MAH. 401.CADDE NO:14/B | 3124959305 | 39.883546,32.875025 | 7 |

```
SELECT * FROM public.boundary
```

Output Messages Notifications

| objectid bigint | adi text | nufus bigint | alan double precision |
|---|---|---|---|
| 20 | NALLIHAN | 29209 | 2064562771.97561 |
| 21 | POLATLI | 121858 | 3417652064.52202 |
| 2 | ALTINDAĞ | 363687 | 158087685.60773 |
| 4 | BALA | 21618 | 1838805217.77147 |
| 7 | ÇANKAYA | 922536 | 405053298.575768 |
| 8 | ÇUBUK | 86055 | 1153958210.78625 |
| 18 | KIZILCAHAMAM | 25179 | 1628363973.99857 |
| 11 | EVREN | 2847 | 218142462.355786 |
| 12 | GÖLBAŞI | 122288 | 1500824326.10754 |
| 10 | ETİMESGUT | 527959 | 149998118.843283 |
| 19 | MAMAK | 607878 | 342333031.698939 |
| 6 | ÇAMLIDERE | 6479 | 753671444.29128 |
| 17 | KEÇİÖREN | 889876 | 154241081.372811 |
| 25 | YENİMAHALLE | 632286 | 420410475.517265 |
| 23 | SİNCAN | 506950 | 877338619.330992 |
| 1 | AKYURT | 30245 | 372164960.912904 |
| 3 | AYAŞ | 12678 | 1148105710.1146 |
| 5 | BEYPAZARI | 47582 | 1642328417.80295 |
| 9 | ELMADAĞ | 43776 | 632285857.778195 |
| 13 | GÜDÜL | 8392 | 515471095.945902 |
| 14 | HAYMANA | 28355 | 2200469299.95409 |
| 15 | KAZAN | 51764 | 590328711.044076 |
| 16 | KALECİK | 13388 | 1115667656.39241 |
| 22 | PURSAKLAR | 133961 | 151221564.220721 |
| 24 | ŞEREFLİKOÇHİSAR | 33729 | 2045185937.56817 |

# Entity-Relationship Diagram





# Nöbetçi Eczaneler Verisinin İlçelerin Merkez Noktalarına Olan Uzaklarını Hesaplama

Query | Query History

```sql
WITH ankara_cankaya AS (
    SELECT
        39.9334 AS center_lat, 32.8597 AS center_lon -- Çankaya'nın merkez koordinatları
)
SELECT
    *,
    -- Merkeze olan uzaklığı haversine formülü ile hesapla
    6371 * acos(
        cos(radians(ankara_cankaya.center_lat)) *
        cos(radians(split_part(loc, ',', 1)::numeric)) *
        cos(radians(split_part(loc, ',', 2)::numeric) - radians(ankara_cankaya.center_lon)) +
        sin(radians(ankara_cankaya.center_lat)) * sin(radians(split_part(loc, ',', 1)::numeric))
    ) AS distance_to_center_km
FROM
    pharmaciess,
    ankara_cankaya
ORDER BY
    distance_to_center_km;
```

Data Output | Messages | Notifications

| | name text | dist text | address text | phone text | loc text | ilce_id integer | center_lat numeric | center_lon numeric | distance_to_center_km double precision |
|---|---|---|---|---|---|---|---|---|---|
| 1 | YENİ BAYRAMOĞLU | ÇANKAYA | MARMARA SOK. NO:20/A SIHHIYE | 3124356969 | 39.926599,32.85885 | 7 | 39.9334 | 32.8597 | 0.7597018666067853 |
| 2 | ROYAL | ÇANKAYA | Bahçelievler Mah.Kazakistan Cad. No:5/D ÇANKAYA/ANKA... | 3122154770 | 39.931859,32.823205 | 7 | 39.9334 | 32.8597 | 3.116432680745195 |
| 3 | YENİ ÇUKURAMBAR | ÇANKAYA | KIZILIRMAK MAH. 1425. CAD. NO:33/A ÇANKAYA | 3122844688 | 39.901246,32.812006 | 7 | 39.9334 | 32.8597 | 5.415512999035471 |
| 4 | MİRAY | ÇANKAYA | ÖVEÇLER MAH. AHMET HASIM CAD. NO:80/A | 3124836321 | 39.88692198,32.83328016 | 7 | 39.9334 | 32.8597 | 5.638023251339118 |
| 5 | DÖRT MEVSİM | ÇANKAYA | BİRLİK MAH. 401.CADDE NO:14/B | 3124959305 | 39.883546,32.875025 | 7 | 39.9334 | 32.8597 | 5.695535691282636 |
| 6 | YEŞİLBAĞ | ÇANKAYA | Mutlukent Mah. 2074. Sok. No:11 | 3122354884 | 39.899314,32.707654 | 7 | 39.9334 | 32.8597 | 13.509734854885764 |
| 7 | YAŞAMKENT MİNE | ÇANKAYA | ALACAATLI MAH. MEHMET RİFAT BÖREKÇİ CAD. NO:35/B-E | 3125024050 | 39.853974,32.647384 | 7 | 39.9334 | 32.8597 | 20.1516729389572 |



# Başvuran Kişi Verilerinin Nöbetçi Eczanelere Olan Uzaklıklarını Hesaplama

Query Query History

```sql
--Başvuran Kişi verilerinin Nöbetçi Eczanelere olan uzaklıkları hesaplanmalıdır

ALTER TABLE basvuru_noktaları
ADD COLUMN location text;

UPDATE basvuru_noktaları
SET location = CASE
    WHEN basvuru_numarasi = 1 THEN '39.895529890677054, 32.85406123627972'
    WHEN basvuru_numarasi = 2 THEN '39.891974665659134, 32.841726726197855'
    WHEN basvuru_numarasi = 3 THEN '39.96048921855266, 32.81581416852985'
    WHEN basvuru_numarasi = 4 THEN '39.902884397819015, 32.86023896852803'
    WHEN basvuru_numarasi = 5 THEN '39.91897491338078, 32.85624068202247'
    ELSE location
END;
SELECT
    basvuru_noktaları.adresi AS person_address,
    pharmaciess.address AS eczane_address,
    6371 * acos(
        cos(radians(CAST(SUBSTRING(basvuru_noktaları.location FROM '^(.*),') AS numeric))) *
        cos(radians(CAST(SUBSTRING(pharmaciess.loc FROM '^(.*),') AS numeric))) *
        cos(radians(CAST(SUBSTRING(pharmaciess.loc FROM ',(.*)$') AS numeric)) - radians(CAST(SUBSTRING(basvuru_noktaları.location FROM ',(.*)$') AS numeric))) +
        sin(radians(CAST(SUBSTRING(basvuru_noktaları.location FROM '^(.*),') AS numeric))) * sin(radians(CAST(SUBSTRING(pharmaciess.loc FROM '^(.*),') AS numeric))
    ) AS distance_km
FROM
    basvuru_noktaları
CROSS JOIN
    pharmaciess;
```



NetcadCase Private

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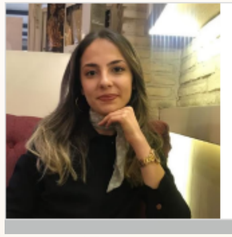
gizemcann Update README.md 137fd31 now 3 commits

| | | |
|---|---|---|
| Project | Netcad CaseStudy added. | 18 hours ago |
| .gitattributes | Initial commit | 18 hours ago |
| README.md | Update README.md | now |

README.md

# NetcadCase

This repository contains case study project from the NETCAD YAZILIM A.Ş.



**gizemcann - Overview**

Geomatics Engineer. gizemcann has 10 repositories available. Follow their code on GitHub.

GitHub

https://github.com/gizemcann

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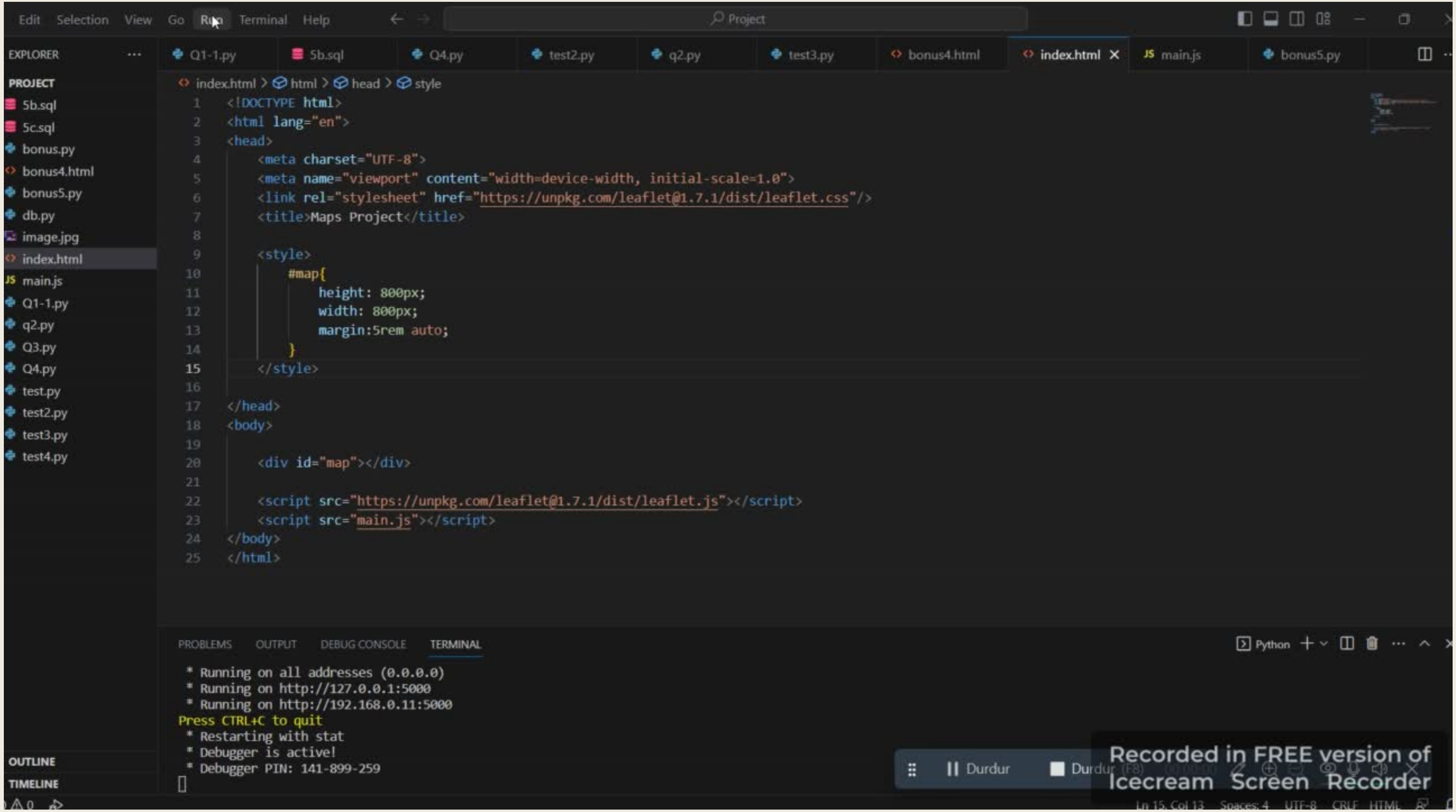
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# Interactive map layers using LeafLets.js and Mapbox API



index.html > html > head > style > #map

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <link rel="stylesheet" href="https://unpkg.com/leaflet@1.7.1/dist/leaflet.css"/>
    <title>Maps Project</title>

    <style>
        #map{
            height: 800px;
            width: 800px;
            margin:5rem auto;
        }
    </style>

</head>
<body>

    <div id="map"></div>

    <script src="https://unpkg.com/leaflet@1.7.1/dist/leaflet.js"></script>
    <script src="main.js"></script>
</body>
</html>
```

main.js > ...

```
const apiKey = 'pk.eyJ1IjoiYWxmcmVkMjAxNiIsImEiOiJja2RoMHkyd2wwdnZjMnJ0MTJwbnVmeng5In0.E4QbAFjiWLY8k3AFhDtErA';

const mymap = L.map('map').setView([40.770116, -73.967909], 13);

L.tileLayer('https://api.mapbox.com/styles/v1/{id}/tiles/{z}/{x}/{y}?access_token={accessToken}', {
    maxZoom: 18,
    id: 'mapbox/streets-v11',
    tileSize: 512,
    zoomOffset: -1,
    accessToken: apiKey
}).addTo(mymap);

// Adding Marker
const marker = L.marker([40.748708, -73.985433]).addTo(mymap);

// Add popup message
let template = `
<h3>Empire State Building</h3>
<div style="text-align:center">
    <img width="150" height="150"src="image.jpg"/>
</div>
`
marker.bindPopup(template);
// Add circle
const circle = L.circle([40.748708, -73.985433], {
    radius:500,
    color: 'green',
    fillColor: 'red',
    fillOpacity:0.2
}).addTo(mymap).bindPopup('I am a circle')

// Add Polygon
const polygon = L.polygon([

    [40.73618, -73.97545],
    [40.731497, -73.974342],
    [40.729324, -73.972234],
    [40.732368, -73.981293],
], {
    color:'blue',
    fillColor:'blue',
    fillOpacity:0.2
}).addTo(mymap).bindPopup(' I am a polygon')
```

# TEŞEKKÜRLER